

AI-POWERED COLLEGE HELPDESK

A Formal Hackathon Project Report

1. Introduction

The increasing use of digital technologies in educational institutions has created a need for efficient and intelligent student-support systems. Students frequently require information related to academic regulations, examinations, attendance, fees, scholarships, hostel facilities, timetables, college notices, and administrative procedures. Traditionally, students depend on faculty members, administrative offices, notice boards, or college websites to obtain this information. This process can be time-consuming and may result in delays or difficulty in finding the correct information.

The AI-Powered College Helpdesk is proposed as an intelligent digital platform designed to provide students with quick, accessible, and reliable assistance. The system uses Artificial Intelligence (AI), Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG) to answer student queries using authorized college documents and information.

The proposed platform acts as a centralized helpdesk where students can ask questions using natural language. Instead of depending only on predefined answers, the system searches relevant institutional documents and generates a contextual response. If the requested information is unavailable, the system can guide the student toward the appropriate department or create a support ticket.

2. Problem Statement

Students in colleges often face difficulties in obtaining accurate and timely information. Important information may be distributed across different documents, notices, departments, websites, and administrative offices. Repeated queries also increase the workload of faculty and administrative staff.

Major problems include:

- Difficulty in locating relevant college information.
- Repeated questions regarding attendance, examinations, fees, and regulations.
- Delays in receiving responses from administrative departments.
- Information being distributed across multiple platforms.
- Difficulty understanding lengthy notices and documents.
- Risk of receiving outdated or incorrect information.
- Lack of a centralized mechanism for unresolved student complaints.

Therefore, there is a need for an intelligent system that can provide information quickly while referring to authorized institutional sources.

3. Objectives

The primary objective of the AI-Powered College Helpdesk is to develop a centralized AI-based support platform for students.

The major objectives are:

1. To provide students with instant answers to common college-related queries.
 2. To use official college documents as the primary knowledge source.
 3. To implement RAG technology to improve the accuracy and relevance of responses.
 4. To provide references or sources for generated answers.
 5. To support natural-language queries, including simple conversational language.
 6. To summarize lengthy college notices and documents.
 7. To provide a ticket-generation mechanism for unresolved issues.
 8. To provide an administrative dashboard for managing documents and student requests.
 9. To reduce repetitive workload for college administrative staff.
 10. To create a scalable platform that can later integrate with existing college ERP systems.
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4. Proposed System

The proposed system consists of two major interfaces: a Student Interface and an Administrator Interface.

Student Interface

Students can access a web-based chatbot and enter questions such as:

- “What is the attendance requirement?”
- “When are the semester examinations?”
- “How can I apply for a scholarship?”

- “What documents are required for hostel admission?”
- “Where can I submit my examination form?”

The system processes the query, searches the relevant institutional knowledge base, and generates an appropriate response.

Administrator Interface

The administrator can:

- Upload college documents.
 - Update or remove outdated information.
 - Manage notices.
 - Monitor frequently asked questions.
 - View unresolved queries.
 - Manage student support tickets.
 - Assign tickets to appropriate departments.
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5. Working Methodology

The system follows a Retrieval-Augmented Generation architecture.

Step 1: Document Collection

Official college documents such as academic regulations, notices, examination guidelines, fee information, hostel rules, and scholarship documents are collected.

Step 2: Document Processing

Uploaded documents are converted into machine-readable text. Large documents are divided into smaller sections or chunks so that relevant information can be efficiently retrieved.

Step 3: Embedding Generation

The processed text is converted into numerical representations called embeddings. These embeddings capture the semantic meaning of the information.

Step 4: Vector Database

The embeddings are stored in a vector database. When a student asks a question, the system searches the database for information that is semantically related to the query.

Step 5: AI Response Generation

The retrieved information is provided to the language model as context. The AI generates a response based on the retrieved institutional information.

Step 6: Source Reference

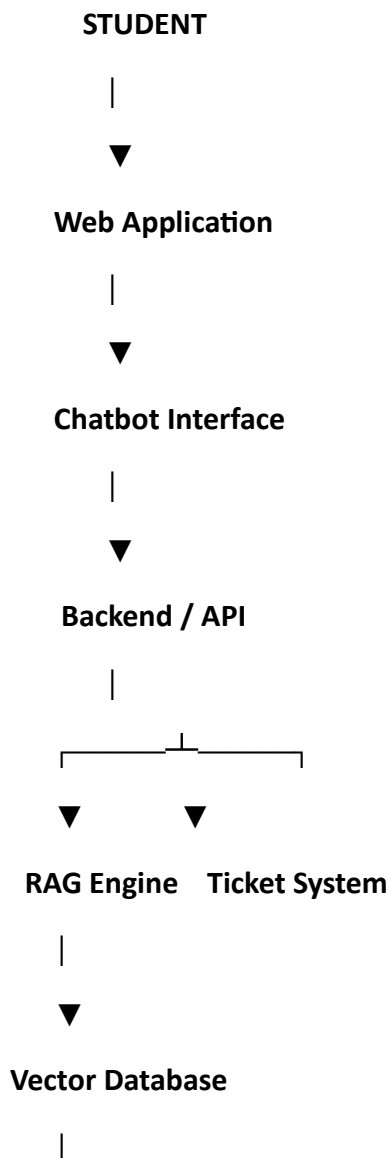
The system can display the relevant document or page reference along with the answer. This allows students to verify important information.

Step 7: Ticket Escalation

If the required information cannot be found, the system can offer the student an option to create a support ticket. The ticket can then be forwarded to the appropriate department.

6. System Architecture

The basic architecture of the proposed system is:





College Documents



AI / LLM



Answer + Source

This architecture separates the user interface, AI processing, institutional knowledge, and administrative functions, making the system easier to maintain and expand.

7. Technology Stack

Component	Suggested Technology
Frontend	React.js, HTML, CSS, Tailwind CSS
Backend	Python, FastAPI
AI	Large Language Model API
AI Technique	Retrieval-Augmented Generation
Embeddings	Embedding Model
Vector Database	FAISS / ChromaDB
Database	PostgreSQL
Document Processing PDF/Text Processing Libraries	
Authentication	JWT / College Login
Deployment	Cloud Hosting

The selected technologies can be modified according to the resources and requirements of the college or hackathon.

8. Key Features

8.1 AI-Based Question Answering

The chatbot allows students to communicate using natural language rather than searching through multiple documents manually.

8.2 Document-Based Knowledge

The AI retrieves information from authorized college documents, helping reduce unsupported responses.

8.3 Source-Based Responses

Important answers can include the document name and relevant page or section, allowing students to verify the information.

8.4 Notice Summarization

The system can summarize lengthy notices into important points such as dates, deadlines, requirements, and instructions.

8.5 Multilingual Interaction

The platform can be extended to support English, Hindi, and Hinglish queries, improving accessibility for students.

8.6 Automated Ticket Management

Queries that cannot be resolved by the AI can be converted into support tickets and forwarded to the relevant department.

8.7 Administrative Dashboard

Administrators can manage the knowledge base, notices, documents, and student support requests through a centralized dashboard.

9. Benefits

The proposed system provides several benefits to students and college administration.

For Students

- Quick access to college information.
- Reduced dependency on physical offices.
- Easy access to information through conversational queries.
- Better understanding of lengthy notices.

- Ability to submit unresolved issues digitally.

For College Administration

- Reduction in repetitive queries.
 - Centralized information management.
 - Better organization of student requests.
 - Faster routing of complaints and queries.
 - Useful analytics about frequently requested information.
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10. Expected Outcome

The expected outcome of the project is a functional web-based AI helpdesk capable of answering common student queries using institutional documents.

The prototype will demonstrate:

1. Student login/interface.
2. AI-powered chatbot.
3. Document upload and processing.
4. RAG-based information retrieval.
5. Source-supported answers.
6. Notice summarization.
7. Support-ticket generation.
8. Administrative dashboard.

The system can serve as a foundation for a larger college digital-support platform.

11. Future Scope

The system can be further enhanced with advanced features such as:

- Voice-based AI assistance.
- WhatsApp or mobile application integration.
- Integration with college ERP systems.
- Personalized student dashboards.

- Automated examination reminders.
 - Attendance-related notifications.
 - AI-based workflow automation.
 - Real-time announcement distribution.
 - Analytics for identifying common student problems.
 - AI agents capable of performing authorized administrative tasks.
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12. Conclusion

The AI-Powered College Helpdesk provides a practical application of Artificial Intelligence, Data Science, Natural Language Processing, and Intelligent Automation in the education sector. By combining a conversational interface with Retrieval-Augmented Generation, the system can help students find relevant institutional information without manually searching through multiple documents.

The addition of source references and ticket escalation provides a structured approach for handling both answerable and unresolved queries. At the same time, the administrative dashboard enables college staff to maintain the information used by the system.

Overall, the proposed project demonstrates how AI can be integrated into college support services to create a more accessible, organized, and efficient information-management system.